

If M is an interval module, then, when viewed as a persistence module of one-parameter persistence modules, every slice $M_{(-,i)}$ is either zero or a single one-parameter interval (see Figure 2). This is a direct consequence of the definition of an interval module as an indecomposable thin persistence module supported on a convex connected subset of $G(m, n)$. Since there is at most one non-zero morphism between two one-parameter interval modules, $\text{id} = (\text{id}, \dots, \text{id})$ is the only barcode basis for an interval module.

► **Proposition 7.** *For an interval module M , $\mathcal{GC}_{\text{id}}(M)$ is a directed path.*

Proof. By construction of $\mathcal{GC}_{\text{id}}(M) = \mathcal{GC}(M)$, the vertices at every height of $\mathcal{GC}(M)$ correspond to the intervals in $M_{(-,i)}$. Since M is an interval module, there is at most one vertex at every height. Since M is indecomposable all these vertices have to be connected. Otherwise Proposition 2 and 4 would imply that M is decomposable. Since there are only edges between vertices at consecutive height levels in a graphcode, $\mathcal{GC}(M)$ is a directed path. ◀

Compressed graphcodes. A disadvantage of graphcodes is that every bar of every barcode is represented by a vertex, leading to a high number of vertices and therefore a large graph. However, in practical situations, the barcodes of consecutive slices often do not differ from each other, and the question is how to compress the information of a graphcode without losing any information. The following definition prepares our approach by relaxing the condition that graphcodes only connect bars on consecutive slices.

► **Definition 8** (generalized Graphcode). *A directed graph $\mathcal{G} = (V, E, c)$ with vertex labels*

$$c: V \rightarrow \{1, \dots, m\} \times \{1, \dots, m+1\} \times \{1, \dots, n\}$$

is called a generalized graphcode if:

1. *for all $v \in V$ with $c(v) = (b, d, h)$, we have $b < d$*
2. *for all $v \in V$ and all $(v, w), (v, w') \in E$, we have $h(w) = h(w')$*
3. *for all $(v, w) \in E$ with $c(v) = (b_1, d_1, h_1)$ and $c(w) = (b_2, d_2, h_2)$, we have $[b_2, d_2] \triangleleft [b_1, d_1]$.*

Note that graphcodes are generalized graphcodes where $h(w) = h(v) + 1$ for all edges $(v, w) \in E$. The condition that all out-neighbors of a vertex are on the same height will be convenient in the next section where we generate a minimal presentation out of a generalized graphcode.

We call a vertex w in a generalized graphcode $\mathcal{G}$ *superfluous* if w has exactly one incoming edge (v, w) , v and w have the same birth and death values (i.e., with $c(v) = (b_1, d_1, h_1)$ and $c(w) = (b_2, d_2, h_2)$, we have $b_1 = b_2$ and $d_1 = d_2$), w has at least one outgoing edge, and v has no further outgoing edge. In that case, writing $(w, x_1), \dots, (w, x_s)$ for the outgoing edges of w , we define $\mathcal{G}'$ as the graph obtained from $\mathcal{G}$ by removing w and all incident edges, and adding the edges $(v, x_1), \dots, (v, x_s)$ instead. It is simple to verify that $\mathcal{G}'$ is a generalized graphcode as well. Moreover, the property of being superfluous does not change when eliminating other superfluous vertices in the graphcode. We call a generalized graphcode $\mathcal{G}'$ a *compression* of a graphcode $\mathcal{G}$ if $\mathcal{G}'$ is obtained from $\mathcal{G}$ by removing superfluous vertices. We call $\mathcal{G}'$ *fully compressed* if it does not contain any superfluous vertex. Figure 3 shows an example.

A generalized graphcode also induces a two-parameter persistence module: For a generalized graphcode $\mathcal{G}$, we define its *expansion* $\text{Exp}(\mathcal{G})$ as follows: if a vertex v with $c(v) = (b, d, h)$ has all its outgoing neighbors $w_1, \dots, w_s$ at height h' with $h' - h \geq 2$, remove all edges (v, w_i) , introduce vertices $v_{h+1}, \dots, v_{h'-1}$ with $c(v_{h+i}) := (b, d, h+i)$ and add edges